Optimization of Heterogeneous Edge AI Inference Pipeline on TDA4VM using C7x DSP

Emre Doğan* and Yavuz Selim Kızıltaş†
*Computer Engineering Department, Hacettepe University, Ankara, Türkiye
Student ID: 504251571
†Graduate School of Science and Engineering, Hacettepe University, Ankara, Türkiye
Student ID: N25223483

***Abstract*—Deploying deep learning models on edge devices requires balancing high inference accuracy with strict power, thermal, and latency constraints. While standard edge CPUs or GPUs offer deployment flexibility, they often fail to meet the stringent power-efficiency requirements of advanced industrial vision applications. The Texas Instruments TDA4VM System-on-Chip (SoC) addresses this challenge through a heterogeneous architecture featuring a C7x DSP paired with a Matrix Multiply Accelerator (MMA). This project proposes an optimized heterogeneous inference pipeline tailored for the TDA4VM SoC in order to maximize throughput and energy efficiency, using the TIDL toolkit for INT8 post-training quantization.**

***Index Terms*—edge AI, TDA4VM, C7x DSP, heterogeneous inference, INT8 quantization, TIDL, MSMC, real-time processing.**

I. Introduction

Deploying deep learning models on edge devices requires balancing high inference accuracy with strict power, thermal, and latency constraints. While standard edge CPUs or GPUs offer deployment flexibility, they often fail to meet the stringent power-efficiency requirements of advanced industrial vision applications. The Texas Instruments TDA4VM System-on-Chip (SoC) addresses this challenge through a heterogeneous architecture featuring a C7x DSP paired with a Matrix Multiply Accelerator (MMA). However, fully utilizing this architecture presents several software engineering challenges, including cross-core synchronization, memory bandwidth bottlenecks between DDR and on-chip SRAM, and accuracy degradation caused by model quantization.

II. Related Work

Deploying deep neural networks on edge computing platforms has been extensively studied to address the trade-off between inference accuracy and hardware constraints. Wang et al. [1] provide a comprehensive survey on edge intelligence, highlighting that while standard CPUs and GPUs offer flexibility, they struggle with the strict power limitations of mobile and automotive systems, necessitating heterogeneous SoC designs.

To bridge the gap between heavy deep learning models and low-power edge processors, post-training quantization (PTQ) and integer-only inference have become industry standards. Jacob et al. [2] introduced a foundational quantization scheme that maps FP32 weights and activations to 8-bit integers, significantly reducing latency and memory bandwidth requirements without substantial accuracy drops. This methodology is directly applicable to the INT8 optimization pipeline executed on the C7x DSP and MMA.

In the context of specific vision tasks, such as Infrared Small Target Detection (IRSTD), achieving high precision often relies on complex feature extraction mechanisms. For instance, Li et al. [3] proposed DNA-Net, a dense nested attention network that achieves state-of-the-art accuracy in detecting dim and small targets. However, such intricate architectures are typically designed for high-end GPUs. This project builds upon these hardware and software foundations by taking computationally intensive, state-of-the-art vision models and mapping them onto the highly optimized, heterogeneous architecture of the TDA4VM.

III. Problem Definition

A. System / Application Description

The target system is an embedded real-time computer vision application deployed on the Texas Instruments TDA4VM SoC, aimed at industrial-grade object detection tasks such as Advanced Driver Assistance Systems (ADAS) and autonomous drone perception. The TDA4VM integrates multiple heterogeneous processing cores—including an ARM Cortex-A72 cluster, C7x DSP with MMA, and C66x DSPs—onto a single low-power SoC. The application ingests camera frames, performs neural network inference to detect and classify objects, and outputs structured results (e.g., bounding boxes and class labels) in real time. The core challenge is that the memory and compute resources of the SoC are tightly constrained: the on-chip MSMC SRAM is limited to 8 MB, the power envelope is restricted to automotive-grade levels, and the system must sustain deterministic, low-latency responses in safety-critical scenarios.

B. Objectives

The primary objectives of this project are:

- **Maximize Inference Throughput:** Achieve real-time inference at or above 30 frames per second (FPS) for the selected object detection model on the TDA4VM's C7x DSP and MMA accelerator.
- **Minimize Latency:** Reduce the end-to-end pipeline latency (from frame capture to detection output) to below

33 ms per frame, meeting the hard deadline imposed by real-time ADAS applications.

- **Optimize Energy Efficiency:** Maximize TOPS/W by exploiting INT8 quantization via TIDL and utilizing the MMA's integer arithmetic capabilities instead of floating-point computation.
- **Preserve Model Accuracy:** Limit the accuracy degradation introduced by INT8 post-training quantization to less than 2% mAP relative to the FP32 baseline on the COCO validation set.
- **Efficient Memory Utilization:** Strategically place high-reuse intermediate tensors in the 8 MB MSMC SRAM while managing low-reuse tensors in DDR to reduce memory bandwidth bottlenecks.

C. Constraints

The following constraints govern the design space of this project:

- **Hardware Constraints:** The on-chip MSMC SRAM is limited to 8 MB, shared across all processing cores. DDR bandwidth is shared and constitutes a potential bottleneck for large tensor transfers. The C7x DSP vector width and MMA dimensions impose fixed tiling requirements on supported operators.
- **Power and Thermal Constraints:** The TDA4VM is designed for automotive-grade deployment with a total SoC power budget of approximately 10–15 W. The inference pipeline must remain within this budget under sustained operation.
- **Toolchain Constraints:** Model deployment relies on the TIDL toolkit and TI Processor SDK, which support a defined subset of neural network operators in INT8 mode. Layers not supported by TIDL must fall back to ARM execution, potentially creating latency bottlenecks.
- **Accuracy Constraints:** The quantized model must maintain an mAP within 2% of the FP32 baseline to remain suitable for safety-relevant perception tasks.
- **Real-Time Constraints:** All pipeline stages must meet deterministic timing requirements; unbounded latency spikes due to synchronization or memory contention are unacceptable.

D. Mathematical Formulation (Optimization Track)

Let $\mathcal{L} = \{l_1, l_2, \ldots, l_N\}$ denote the set of N layers in the neural network. Each layer l_i is assigned to a processing core $c_i \in \mathcal{C} = \{\text{ARM, C7x, C66x}\}$. Let $T_i(c_i)$ denote the execution latency of layer l_i on core c_i, and let $E_i(c_i)$ denote its energy consumption. The memory footprint of the activation tensor produced by layer l_i is denoted m_i, and each tensor is allocated either to MSMC SRAM or DDR, encoded by the binary variable $s_i \in \{0, 1\}$, where $s_i = 1$ indicates MSMC placement.

The optimization objective is to minimize total pipeline latency while maximizing energy efficiency:

$$\min_{\{c_i\},\{s_i\}} \quad \sum_{i=1}^{N} T_i(c_i) + \alpha \sum_{i=1}^{N} E_i(c_i) \tag{1}$$

subject to the following constraints:

$$\sum_{i:\, s_i=1} m_i \leq M_{\text{MSMC}} \tag{2}$$

$$\text{mAP(INT8)} \geq \text{mAP(FP32)} - \delta \tag{3}$$

$$T_{\text{pipeline}} \leq T_{\text{deadline}} \tag{4}$$

$$c_i \in \mathcal{C}, \quad s_i \in \{0, 1\}, \quad \forall\, i \tag{5}$$

where $M_{\text{MSMC}} = 8\,\text{MB}$ is the on-chip SRAM capacity (constraint 2), $\delta = 0.02$ is the maximum allowable accuracy drop from INT8 quantization (constraint 3), $T_{\text{deadline}} = 33\,\text{ms}$ is the real-time latency budget (constraint 4), and α is a weighting factor balancing latency and energy objectives.

IV. System Architecture and Design

A. Hardware/Software Architecture

The proposed system is designed as an end-to-end partially asynchronous data pipeline. The major components of the system are listed below:

- **Data Ingestion and Pre-processing Unit (ARM Cortex-A72):** Responsible for capturing input frames from the sensor interface and preparing the data layout required by the neural network (e.g., converting from NHWC to NCHW format).
- **TIDL Execution Engine (C7x DSP + MMA):** Executes the compiled INT8 neural network layers using the optimized inference kernels provided by the TIDL framework.
- **Memory Management Unit:** Utilizes the Multicore Shared Memory Controller (MSMC) to cache intermediate feature maps, thereby reducing costly accesses to external DDR memory. DDR-to-MSMC transfers are managed by the EDMA controller using asynchronous pre-fetching. Intermediate tensors with high access frequency are pinned to MSMC SRAM, while low-reuse tensors remain in DDR to conserve the limited 8 MB on-chip capacity. This placement policy is determined empirically during the profiling phase.
- **Post-processing Unit (C66x DSP / ARM):** Processes the network outputs and converts them into actionable outputs such as bounding boxes, object labels, or detection metrics.

B. Dataflow and System Model

The core optimization strategy relies on workload distribution across heterogeneous processing units. Compute-intensive tensor operations will be offloaded to the C7x DSP together with the MMA accelerator, while sequential tasks such as image pre-processing and post-processing will be executed on the ARM Cortex-A72 or the C66x DSP cores.

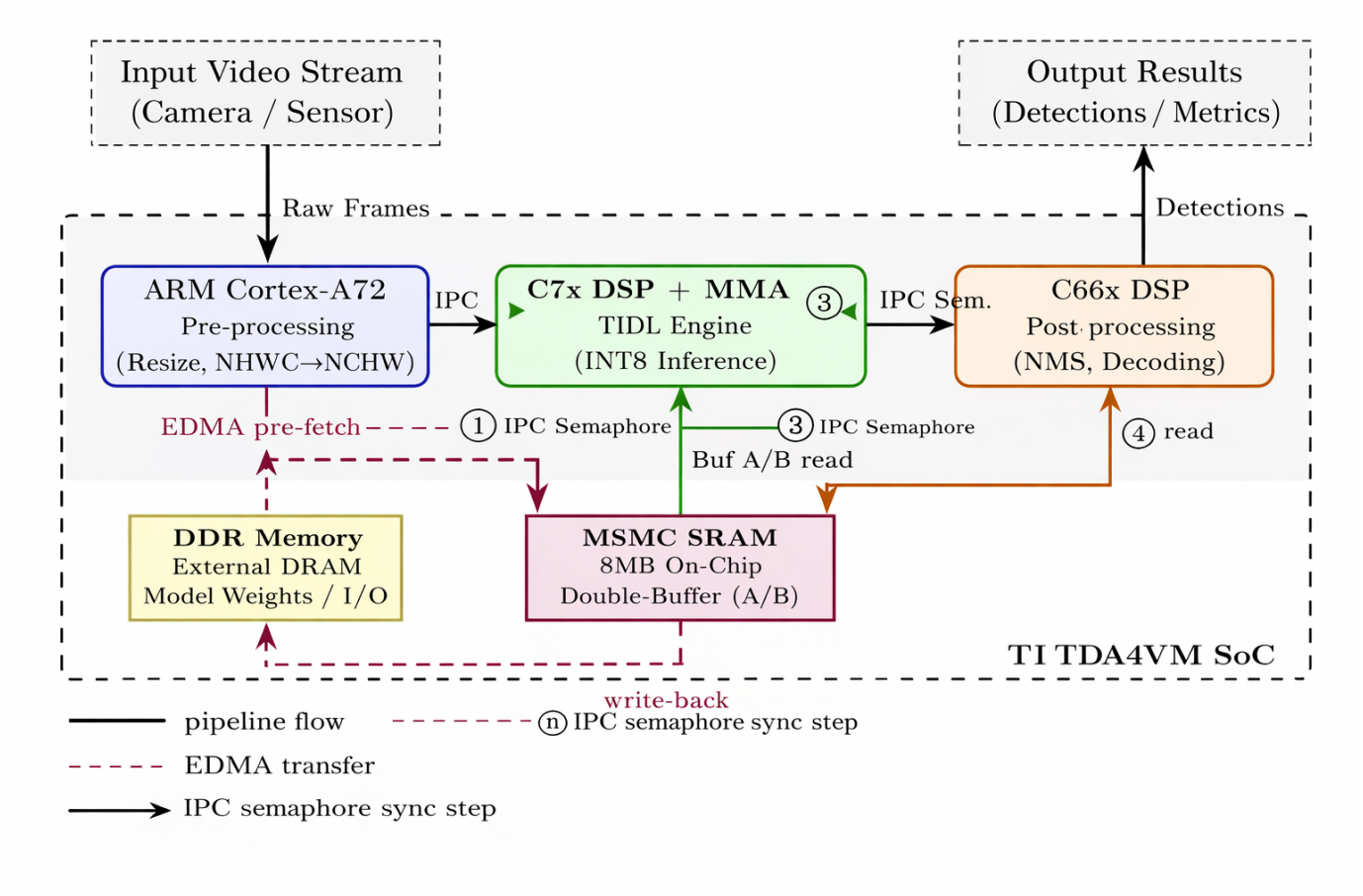

Fig. 1. Heterogeneous Edge AI inference pipeline on the TI TDA4VM SoC.

Task partitioning decisions are made *statically* at compile time, based on the computational profile of each pipeline stage obtained during baseline profiling. Compute-bound layers such as convolutions and matrix multiplications are assigned to the C7x DSP and MMA, while memory-bound or control-flow-heavy tasks such as NMS decoding and format conversion are assigned to the ARM or C66x cores. This assignment is validated using TI SysBIOS profiling utilities; if runtime load imbalance is observed, an adaptive round-robin policy on the C66x cores will be explored.

The pipeline operates in a *partially asynchronous* manner. Inter-stage communication relies on *double-buffering* through shared MSMC SRAM regions, so that one buffer is written by the upstream stage while the downstream stage reads from the other simultaneously. Synchronization between cores is achieved via hardware semaphores provided by the TI SysBIOS IPC framework, which avoids costly busy-wait polling. Data movement between DDR and MSMC SRAM is handled by the EDMA controller: DMA transfers are issued asynchronously so that memory pre-fetching of the next input tile overlaps with the C7x DSP processing the current tile, minimising pipeline stalls due to memory latency.

C. Requirements Analysis (Implementation Track)

1) Functional Requirements:

- The system shall accept raw image frames from the camera sensor interface and perform preprocessing (resizing, normalization, and format conversion) on the ARM Cortex-A72 core.
- The system shall execute the INT8-quantized neural network model on the C7x DSP and MMA using the TIDL inference engine.
- The system shall produce per-frame detection outputs including bounding box coordinates, class identifiers, and confidence scores.
- The system shall support multiple candidate backbone models (YOLOX-Nano, YOLOX-Tiny, SSD-MobileNetV2, EfficientDet-Lite0) and allow model selection at deployment time.
- The system shall export profiling data (per-stage latency, MAC utilization, and memory bandwidth) to enable post-hoc analysis and further optimization.

2) Non-Functional Requirements:

- **Performance:** The system shall sustain a minimum throughput of 30 FPS with an end-to-end latency not exceeding 33 ms per frame.

- **Accuracy:** The mAP of the deployed INT8 model shall remain within 2% of the FP32 baseline on the COCO validation set.
- **Power Consumption:** The inference pipeline shall operate within the TDA4VM automotive-grade power envelope ($\leq$15 W total SoC).
- **Memory Footprint:** The combined activation and weight memory allocated to MSMC SRAM shall not exceed the 8 MB on-chip capacity.
- **Determinism:** The pipeline shall guarantee bounded worst-case execution time per frame with no unbounded blocking operations at runtime.
- **Portability:** The software stack shall be structured to allow model swapping without changes to the pipeline orchestration layer.

D. Timing and Execution Model

The pipeline is governed by a periodic frame-triggered execution model with a nominal frame period of $T_f = 33$ ms (30 FPS). Each frame triggers a cascade of dependent tasks across heterogeneous cores, as illustrated in Fig. 1. The end-to-end latency budget is decomposed across pipeline stages as follows: preprocessing on the ARM core is allocated 2–3 ms; TIDL inference on the C7x DSP and MMA is targeted at 20–25 ms; and post-processing (NMS decoding and output formatting) on the C66x or ARM is allocated 3–5 ms, leaving a margin for synchronization and DMA transfer overhead.

Inter-stage synchronization relies on hardware semaphores managed by the TI SysBIOS IPC framework, ensuring that downstream stages are not scheduled until upstream buffers are committed. The EDMA controller issues asynchronous DMA transfers to overlap DDR-to-MSMC data movement with C7x computation, preventing memory stalls from appearing on the critical path. Worst-case execution times for each stage will be measured empirically using TI performance counters during the profiling phase (Milestone 4), and the static task assignment will be revised if any stage consistently exceeds its budget.

E. Model of Computation

The system follows a *synchronous dataflow* (SDF) model at the inter-core level, where each pipeline stage consumes and produces a fixed number of tokens (frames or tiles) per firing. Within the TIDL inference stage, the C7x DSP processes the network in a *layer-sequential* order, tiling large feature maps to fit within the MSMC SRAM budget when necessary. The double-buffering scheme between the preprocessing and inference stages introduces one frame of pipeline latency but enables fully overlapped execution of consecutive frames, effectively decoupling producer and consumer rates at the inter-core boundary.

At the intra-stage level, the C7x DSP's vector execution model follows an *SIMD dataflow* pattern: the MMA processes fixed-size matrix tiles in a systolic manner, while the C7x scalar pipeline handles activation functions and non-linearity operations concurrently. This two-level model of computation—SDF at the system level and SIMD dataflow at the accelerator level—allows the execution schedule to be analyzed statically, supporting the determinism requirement.

F. Assumptions and Design Limitations

The following assumptions and limitations apply to this project:

- **Static Workload:** The neural network graph topology and operator types are fixed at compile time. Dynamic control flow (e.g., conditional branches inside the network) is not supported by the TIDL toolchain and is not considered.
- **Single-Model Inference:** The pipeline is designed to execute a single detection model at a time. Multi-model concurrent execution is outside the scope of this project.
- **Calibration Data Availability:** INT8 post-training quantization via TIDL requires a representative calibration dataset. It is assumed that a suitable subset of the COCO dataset is available for this purpose.
- **Operator Coverage:** Not all neural network operators are supported in INT8 mode by the TIDL runtime. Unsupported layers will fall back to ARM Cortex-A72 floating-point execution, which may introduce latency bottlenecks for architectures with non-standard operators.
- **Thermal Steady State:** Power and energy measurements are taken under sustained inference load at thermal steady state, as transient thermal effects may cause dynamic frequency scaling that distorts single-frame measurements.

G. Resource Model

This project proposes an optimized heterogeneous inference pipeline tailored for the TDA4VM SoC in order to maximize throughput and energy efficiency. The methodology focuses on migrating a vision-based deep learning model (e.g., a lightweight CNN for object detection) to the C7x DSP. The Texas Instruments Deep Learning (TIDL) toolkit will be used to perform Post-Training Quantization (PTQ), converting the baseline FP32 model to an INT8 format. Table I summarises the candidate models considered for deployment, comparing their estimated activation memory footprint against the 8 MB MSMC SRAM budget and their reported COCO mAP accuracy.

TABLE I
CANDIDATE MODEL QUICK-REFERENCE: ACTIVATION MEMORY VS. MSMC SRAM FIT AND ACCURACY.

Model	Input Size	Est. Act. Memory	SRAM Fit	mAP (COCO)
YOLOX-Nano	416×416	∼2–3 MB	Full	∼25
YOLOX-Tiny	416×416	∼3–4 MB	Full	∼32
YOLOX-S	640×640	∼6–8 MB	Marginal	∼40
SSD-MobileNetV2	300×300	∼3 MB	Full	∼22
EfficientDet-Lite0	320×320	∼3–4 MB	Full	∼26
EfficientDet-Lite1	384×384	∼5–6 MB	Likely	∼31
PeleeNet-SSD	304×304	∼3–4 MB	Full	∼22

V. Methodology

A. Implementation Details (Implementation Track)

- **Milestone 1 (Weeks 1–2):** Baseline model selection, FP32 training and validation, and development environment setup (Processor SDK, Code Composer Studio, and TI Edge AI Studio). Optimization objectives, decision variables, and evaluation metrics (latency, FPS, TOPS/W, mAP) will be formally defined at this stage.
- **Milestone 2 (Weeks 3–4):** Model export to ONNX format and preliminary benchmarking using TI Edge AI Studio to obtain estimated latency, throughput, and memory footprint figures for each candidate model without requiring on-board deployment. Models that fail to meet the latency or SRAM fit targets at this stage will be eliminated from further consideration. Surviving models will proceed to TIDL model import, INT8 quantization, and accuracy evaluation using the PC host emulator.
- **Milestone 3 (Weeks 5–6):** Implementation of the heterogeneous inference pipeline on the target hardware and distribution of tasks across ARM, C66x, and C7x cores. MSMC double-buffering and EDMA-based asynchronous data transfer scheduling will be configured, and IPC semaphore synchronization will be implemented at this stage.
- **Milestone 4 (Weeks 7–8):** System profiling using performance counters, benchmarking (latency, FPS, energy efficiency, and MAC utilization), and preparation of the final project report. Memory placement will be optimised based on profiling data; per-stage latency and bandwidth will be measured; and latency-energy trade-offs will be analysed by comparing the optimised system against the baseline.

B. Optimization Pipeline (Optimization Track)

The optimization pipeline proceeds in four sequential phases. In the **first phase** (model preparation), a pre-trained FP32 model is selected from the candidate set in Table I and exported to the ONNX interchange format. In the **second phase** (quantization), the TIDL model import tool ingests the ONNX graph and performs layer-wise INT8 post-training quantization using a representative calibration subset of the COCO dataset. Per-layer scale factors and zero points are computed via min-max calibration, and quantization-sensitive layers identified by accuracy profiling may be retained in higher precision. In the **third phase** (memory placement optimization), the activation tensor memory footprint of each layer is profiled on the host emulator. Tensors are ranked by access frequency, and a greedy bin-packing algorithm is used to assign the highest-reuse tensors to MSMC SRAM within the 8 MB budget, with the remainder mapped to DDR. In the **fourth phase** (pipeline tuning), EDMA transfer sizes and double-buffer depths are swept empirically to identify configurations that maximize DDR-to-MSMC prefetch overlap with C7x execution, minimizing pipeline stall cycles.

C. Algorithm Description

Algorithm 1 summarises the MSMC tensor placement policy applied during the memory optimization phase. For each layer in the network, the activation tensor is characterized by its size m_i and its estimated reuse count r_i, which is derived from the layer's position in the dataflow graph and the number of downstream layers that consume its output. Tensors are sorted in descending order of reuse count and greedily assigned to MSMC SRAM until the 8 MB capacity M_{MSMC} is exhausted; remaining tensors are mapped to DDR. This greedy policy approximates the optimal solution to the SRAM assignment sub-problem (constraint 2) in linear time, making it practical for deployment-time compilation.

Input: Layer list $\mathcal{L}$, activation sizes $\{m_i\}$, reuse counts $\{r_i\}$, capacity M_{MSMC}
Output: Placement assignment $\{s_i\} \in \{0,1\}^N$
Sort $\mathcal{L}$ by r_i in descending order
used $\leftarrow 0$
for each layer l_i in sorted $\mathcal{L}$ **do**
  if used $+ m_i \leq M_{\text{MSMC}}$ **then**
    $s_i \leftarrow 1$ {Place in MSMC SRAM}
    used $\leftarrow$ used $+ m_i$
  else
    $s_i \leftarrow 0$ {Place in DDR}
  end if
end for
return $\{s_i\}$

VI. Experimental Setup

A. Platform and Tools

All experiments will be conducted on the Texas Instruments TDA4VM Evaluation Module (EVM). The software stack consists of the TI Processor SDK for TDA4VM (RTOS variant), which includes the TIDL runtime, the SysBIOS real-time operating system, the EDMA driver, and the IPC framework. Model preparation and INT8 quantization are performed on a host PC (Ubuntu 22.04, x86-64) using the TIDL import tool and ONNX Runtime for FP32 baseline evaluation. Code Composer Studio (CCS) is used for cross-compilation, on-target debugging, and hardware performance counter access. System-level profiling utilizes TI's UIA (Unified Instrumentation Architecture) framework to capture per-core execution traces and EDMA transfer logs.

Prior to on-board implementation, candidate models are evaluated using TI Edge AI Studio, a cloud-based development environment that enables model compilation, INT8 quantization, and benchmark estimation (latency, throughput, and memory footprint) for TI SoC targets without requiring physical hardware access. This step serves as an early-stage screening tool: models that fail to meet the latency or SRAM fit targets in the Edge AI Studio simulator are eliminated before committing to the more time-intensive on-device deployment workflow, reducing iteration cost during model selection.

B. Datasets / Workloads

Model accuracy is evaluated on the COCO 2017 validation set (5,000 images), using the standard COCO mAP metric (averaged over IoU thresholds from 0.50 to 0.95). A calibration subset of 500 images sampled from the COCO 2017 training set is used to compute INT8 quantization scale factors via the TIDL calibration pipeline. For pipeline throughput and latency benchmarking, a synthetic workload of 1,000 randomly generated 640×640 RGB frames is used to avoid I/O bottlenecks and ensure that reported FPS and latency figures reflect pure inference performance. Memory bandwidth stress tests are conducted by streaming representative COCO validation images continuously for 60 seconds at maximum throughput to measure sustained DDR bandwidth utilization.

C. Evaluation Metrics

The following metrics will be used to evaluate the proposed system:

- **End-to-End Latency (ms):** The wall-clock time from frame capture to final detection output, measured as the average and 99th-percentile over the benchmark workload.
- **Throughput (FPS):** Sustained frames processed per second under continuous inference load.
- **Energy Efficiency (TOPS/W):** Effective integer operations per second normalized by total SoC power draw, estimated from TDA4VM power monitoring registers.
- **Detection Accuracy (mAP):** Standard COCO mAP evaluated on the COCO 2017 validation set for both the FP32 baseline and the INT8-quantized model.
- **MSMC Bandwidth Savings:** Reduction in DDR read traffic (GB/s) achieved by the MSMC tensor placement policy, measured using hardware performance counters.
- **MAC Utilization (%):** Fraction of theoretical peak MMA throughput achieved during sustained inference, reported per model.

VII. System-Level Considerations

Edge AI is vital for modern autonomous systems, but running accurate deep learning models on small edge devices is difficult due to strict power and heat limits. Solving this optimization problem on the TI TDA4VM chip is not just a technical win; it directly improves public safety, saves energy, and lowers costs.

Safety and Reliability: In real-world applications like self-driving cars (ADAS) or rescue drones, fast and local data processing saves lives. For example, detecting a pedestrian in low light requires quick and accurate vision models. If the system is slow, it can cause fatal accidents. By optimizing the software to run efficiently on the C7x DSP, this project guarantees fast, real-time reactions, making autonomous vehicles much safer.

Sustainability and Cost Reduction: Traditional graphics cards (GPUs) use too much power, draining the batteries of electric cars and drones quickly. By shifting the heavy AI calculations to the highly efficient C7x DSP using INT8 quantization, we drastically cut down energy use. This efficiency also lowers hardware costs, allowing standard, affordable cars to have advanced safety features that are usually limited to luxury vehicles.

Beneficiaries and Impact Measurement: The main beneficiaries are everyday drivers, pedestrians, and emergency teams who rely on safe technology. We will measure this project's success using three clear metrics:

- **Safety:** Response time (Latency in milliseconds) and speed (Frames Per Second).
- **Sustainability:** Energy efficiency measured in performance per watt (TOPS/W).
- **Reliability:** The model's accuracy (e.g., mAP) after the optimization process.

The following design considerations are explicitly addressed as evaluation dimensions of this project. Memory bandwidth limitations will be measured using TI hardware performance counters to quantify the improvement achieved by MSMC placement over DDR. The latency-throughput trade-off will be analysed by varying pipeline buffer depth and reporting both single-frame latency and sustained FPS. Synchronization overhead between ARM, C7x, and C66x cores will be profiled to verify that asynchronous scheduling eliminates bottlenecks. The impact of INT8 quantization on accuracy relative to the FP32 baseline will be reported as an accuracy-latency curve. Energy efficiency in TOPS/W will be estimated from TDA4VM power monitoring registers. Finally, scalability will be evaluated by testing the pipeline at different input resolutions.

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